

COMPOSITE MATERIAL DATASHEETS CFRP IMS/977-2

MATERIAL IDENTIFICATION

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	IMS/977-2-UD
Commercial name	Cytec Cycom 977-2-34%-24KIMS-194
Specifications	Airbus: ABS5321A / AIMS05-01-002A
Description	Carbon fiber tape intermediate modulus (IMS-24K) impregnated in epoxy resin. Service temperature: -55 to 95°C. Cure: 120-135 minutes at 185°C.
Version	1.5 (*) Add clarification about limited validity of the CAI model

REFERENCES

- TEA-T-NT-200078 Rev. C "DERIVATION OF MATERIAL DESIGN VALUES FOR CFRP UD-TAPE PRE-PREG CYTEC IMS/977-2"

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.184 mm	Coefficients of Thermal Expansion (1/°C)	Range	-55..20°C	20..70°C	70..120°C
Density	1660 kg/m ³		α_1	-5.5E-7	-1.7E-6	-1.7E-6
Tg-onset (wet)	133 °C		α_2	3.0E-5	3.3E-5	3.8E-5
Max. Operational Limit (MOL)	100 °C		α_3	3.0E-5	3.3E-5	3.8E-5

ELASTIC MODULI

Property	E_1	E_2	G_{12}	ν_{12}	K_B
Units	MPa	MPa	MPa	-	-
Values	157000	8500	4200	0.35	0.812

Property	E_3	G_{13}	G_{23}	ν_{13}	ν_{23}
Units	MPa	MPa	MPa	-	-
Values	8500	4200	2500	0.35	0.683

Remarks:

- Consolidated design values by Airbus Commercial proposed for design on any condition below MOL, for commonality and convenience
- The K_B is the buckling correction factor for moduli to apply to E_1 , E_2 and G_{12} in buckling stability analyses

PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
F_{1t}	2786	2577	2354	-	0.897	0.925	0.845	-	2499	2312	2112	-
F_{1c}	1602	1265	1086	-	0.827	0.789	0.678	-	1325	1046	897.8	-
F_{2t}	86.4	50.8	47.1	-	0.839	0.588	0.545	-	72.5	42.6	39.5	-
F_{2c}	212.8	153.3	151.9	-	0.9	0.721	0.714	-	191.5	138	136.7	-
F_{12}	111.9	85	84	67.7	0.9	0.759	0.751	0.605	100.7	76.5	75.6	60.9

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	Mean Values ($\mu\epsilon$)				B-basis K_B	E-KDF			B-basis Values ($\mu\epsilon$)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
ϵ_{tall}	14500	13800	13800	-	0.876	0.952	0.952	-	12700	12100	12100	-
ϵ_{call}	10200	8100	6900	-	0.827	0.794	0.676	-	8400	6700	5700	-

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INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis K_B	E-KDF			B-basis Values			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
F_{13} (MPa)	106.1	71.2	61.9	51.9	0.891	0.67	0.583	0.489	94.6	63.4	55.2	46.2
F_{3t} (MPa)	-	-	-	-	-	-	-	-	-	-	-	-
G_{IC} (N/mm)	0.501	-	-	-	0.76	-	-	-	-	-	-	-
G_{IIC} (N/mm)	0.738	-	-	-	0.85	-	-	-	-	-	-	-

No H/W data available for fracture toughness. RT/W mean values reported at QTR are $G_{IC} = 0.382\text{N/mm}$, $G_{IIC} = 0.657\text{N/mm}$

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	RT/Dry	70°C/W	90°C/W	120°C/W		70°C/W	90°C/W	120°C/W	RT/Dry	70°C/W	90°C/W	120°C/W
OHT_{QI}	493.3	493.3	493.3	-	0.9	1	1	-	444	444	444	-
OHC_{QI}	322.1	245.9	234.7	-	0.9	0.763	0.729	-	289.9	221.3	211.2	-
FHT_{QI}	444.1	425.2	436	-	0.9	0.957	0.982	-	399.7	382.6	382.6	-
FHC_{QI}	341.8	313.6	303.1	-	0.9	0.917	0.887	-	307.6	282.2	272.8	-
F_{BR-QI}	951.4	793.4	725.2	686.5	0.886	0.834	0.762	0.722	843	703.1	642.7	608.3

Pull-out allowable force vs bolts tension: $T_{po} = \tau'_{po} \cdot \pi \cdot D_h \cdot t$

...where $\tau'_{po} = EKDF \cdot K_B \cdot F_{13} \cdot \Psi(t/d)$, with: $\Psi = 1 + 0.6 \cdot (t/d - 1)$ if $t/d < 1$, otherwise $\Psi = 1$,
 D_h is the head/nut diameter, d the fastener diameter, and t the laminate thickness

COMPOSITE DAMAGE TOLERANCE

- Residual strength after transverse impact

The following TAI, CAI model is valid for laminates following the standard design rules (symmetric, balanced, with minimum 10% of plies at angles 0°, 90°, 45° and -45°). These design values shall be considered preliminary and not used for structural justification assessments in certification/qualification processes (further testing required to validate this approach).

Property	B-basis Value for all conditions
TAI ($\mu\epsilon$)	6100

$$E/t^2 = A \cdot (1 - e^{-B \cdot d_0/t})$$

Property	Design Value
A (J/mm ²)	5.5
B	3.0

Approximation model type: CDTM-T05

$$CAI = K_B \cdot EKDF \cdot \{ CAI_{\infty-QI} \cdot (3 / K_{TOH})^{C_x} + A / e^{B \cdot e \cdot (E/t^2)} \}$$

Property	Mean Value RT/Dry	B-basis K_B	E-KDF		
			70°C/W	90°C/W	120°C/W
$CAI_{\infty-QI}$ ($\mu\epsilon$)	3000	0.90	0.919		
A ($\mu\epsilon$)	5500				
B_e	1.5				
C_x	1.4				

...where:

- E is the impact energy, not higher than the cut-off identified during damage threats analysis (for many parts of the airframe, 35J as cut-off value shall be acceptable)
- t the laminate thickness
- d_0 the dent depth after impact (typ. 2.0-2.5mm for BVID detectability under a GVI inspection)

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- K_{TOH} is the Stress Concentration Factor for an infinite plate with a circular hole in by-pass load, calculated from laminate apparent moduli for X direction (*) as $K_{TOH} = 1 + \{ 2 \cdot [(E_x/E_y)^{0.5} - \nu_{xy}] + E_x/G_{xy} \}^{0.5}$

(*) CAI shall be calculated not only for the 0° direction (X parallel to material 0° reference), but also for the rest of ply orientations, i.e. the 45° and 90° directions, by rotating the X axis and re-evaluating previous expressions with the new values of E_x , E_y , G_{xy} and ν_{xy} . In a general case CAI strength shall be expressed as 3 scalar values with the allowable strains applicable to plies at 0° (CAI_0), at ±45° (CAI_{45}) and at 90° (CAI_{90}).

- Residual strength after transverse impact near lightening holes (or curved edges)

Airbus DS experience on mechanical strength tests performed on composite panels with holes (including impact damages near the edge)(**), is that the residual strength after impact of that detail, monitored by the tangential strain at the hole edge, is above twice the value of the conventional CAI strength of a laminated plate including a full penetrating impact damage. For this material, a cut-off of 4900µε (2·0.9·0.919·3000µε) is selected as preliminary design value.

Property	B-basis value up to 70°C/W
CAI_{hole} (µε)	4900

(**) TEA-CS-NT-190030 Is. A "Test on spar web panels. Analysis of results."

NT-A0-ECE-050 Ed. A "Criterio de fallo de herrajes del borde de salida del cajon central"

- Residual strength after edge impact

No design data for CAI_E still available